

Digital Grid

From power line inspection to management

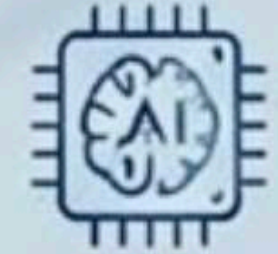
We offer not just drones or AI, but a unified intelligent platform for power line management.



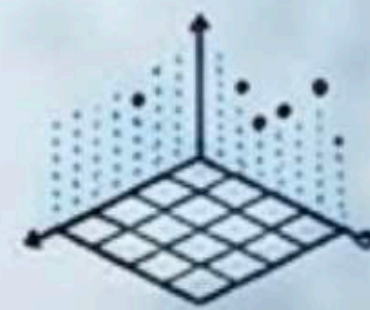
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Autonomous inspections



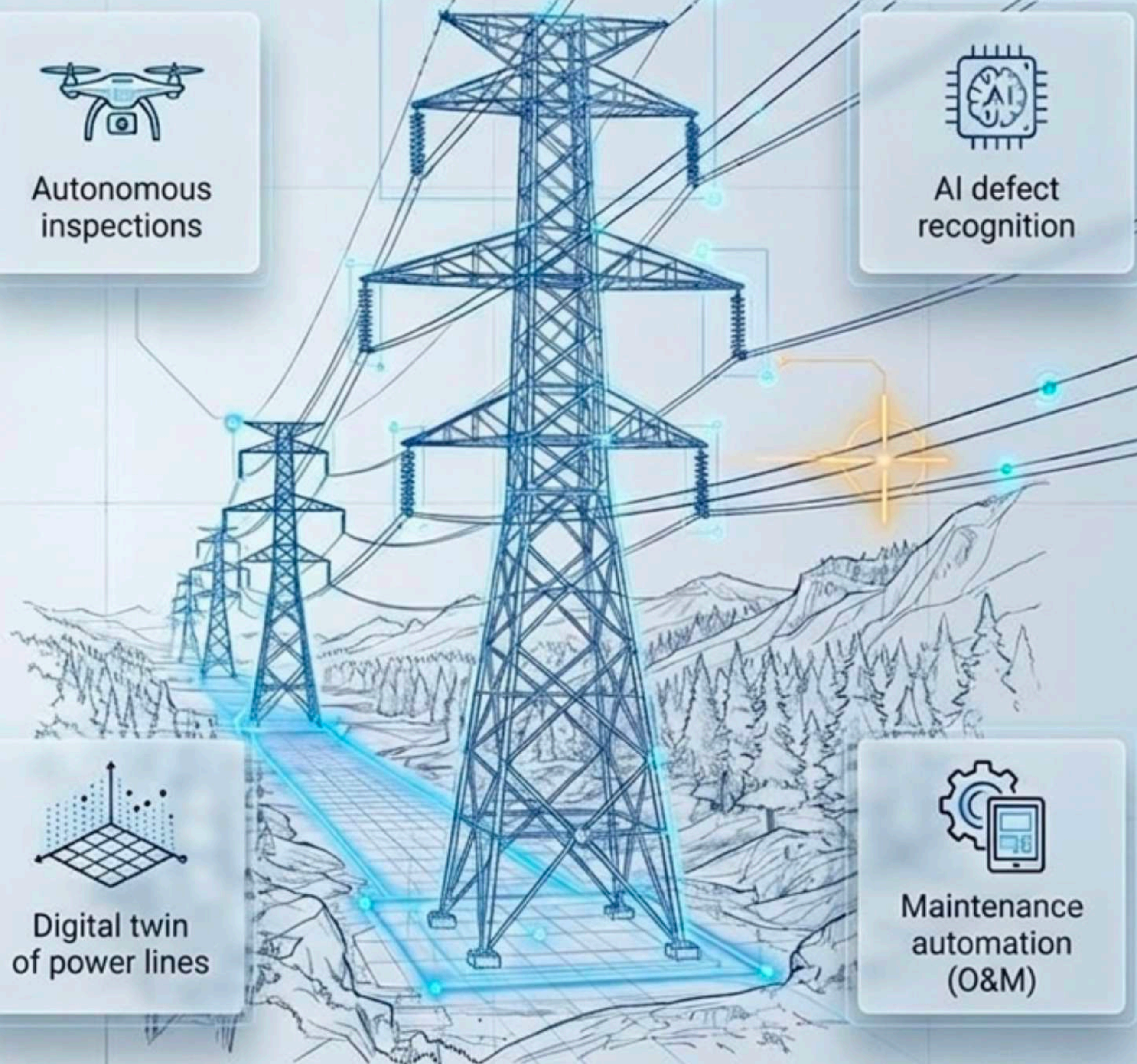
AI defect recognition



Digital twin of power lines



Maintenance automation (O&M)



Current limitations of power line inspections

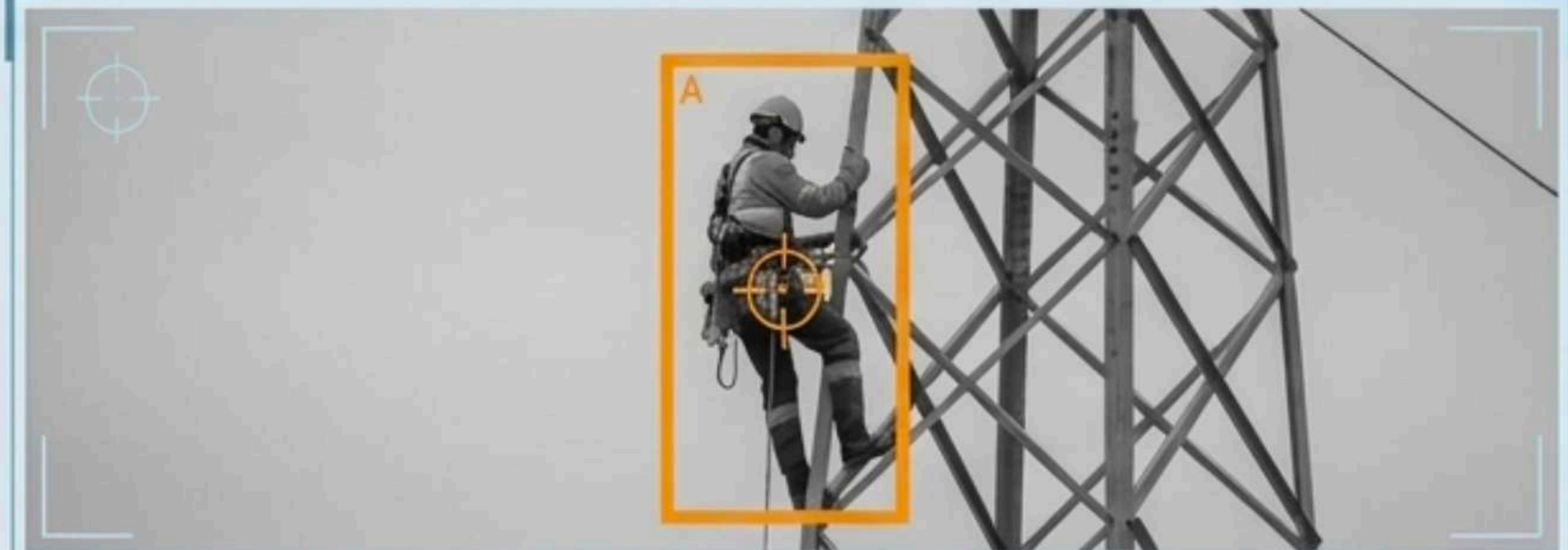


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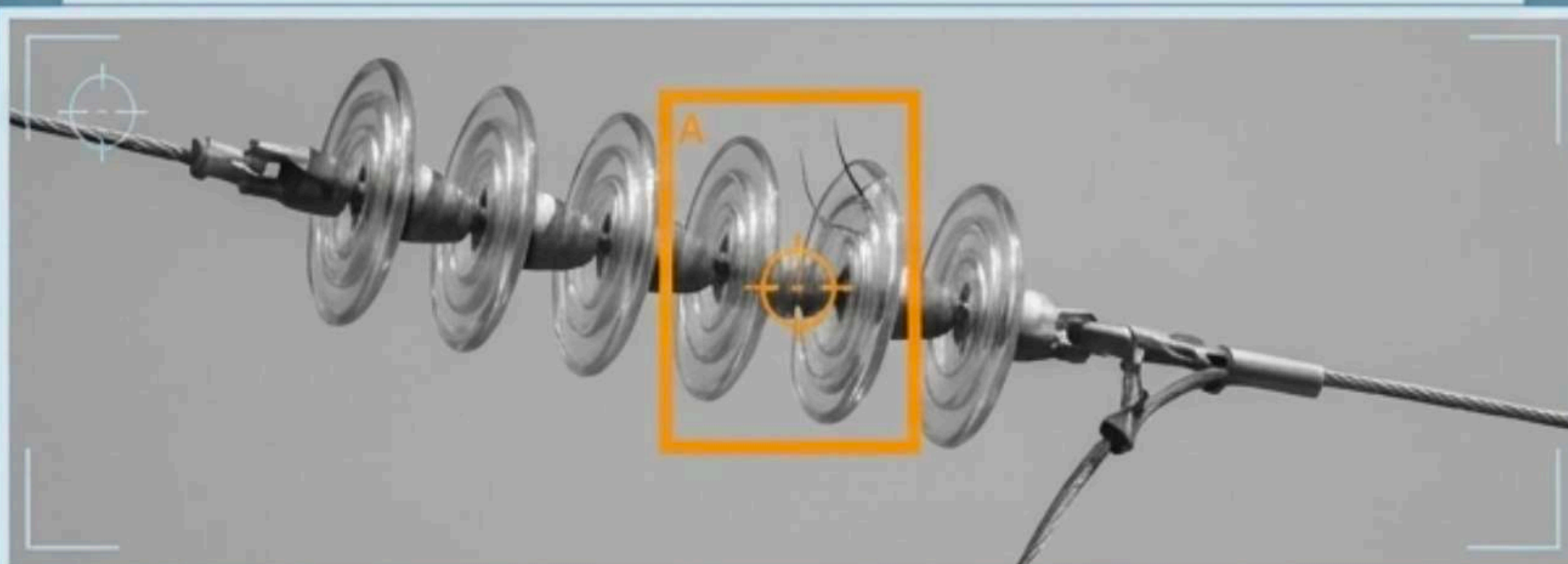
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Manual patrols and high reliance on personnel



Risks for employees



Slow defect detection



Data loss and lack of a unified system

Intelligent platform for power lines



Drones
(inspection)

AI and Data platform
(analysis and storage)

O&M
(task management)

Fully autonomous inspection



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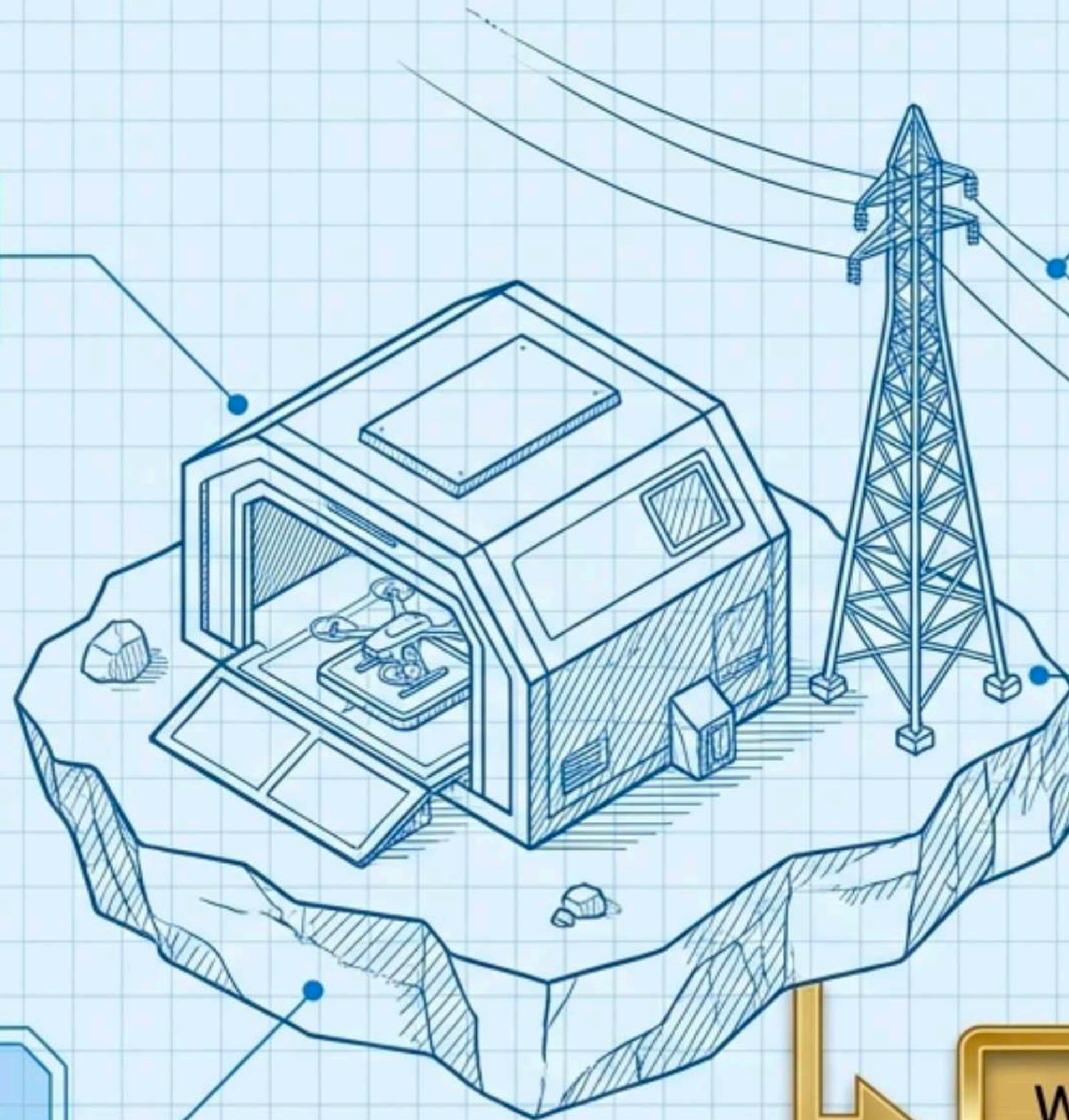
Auto-launch and
auto-landing of drones

Flight without
an operator

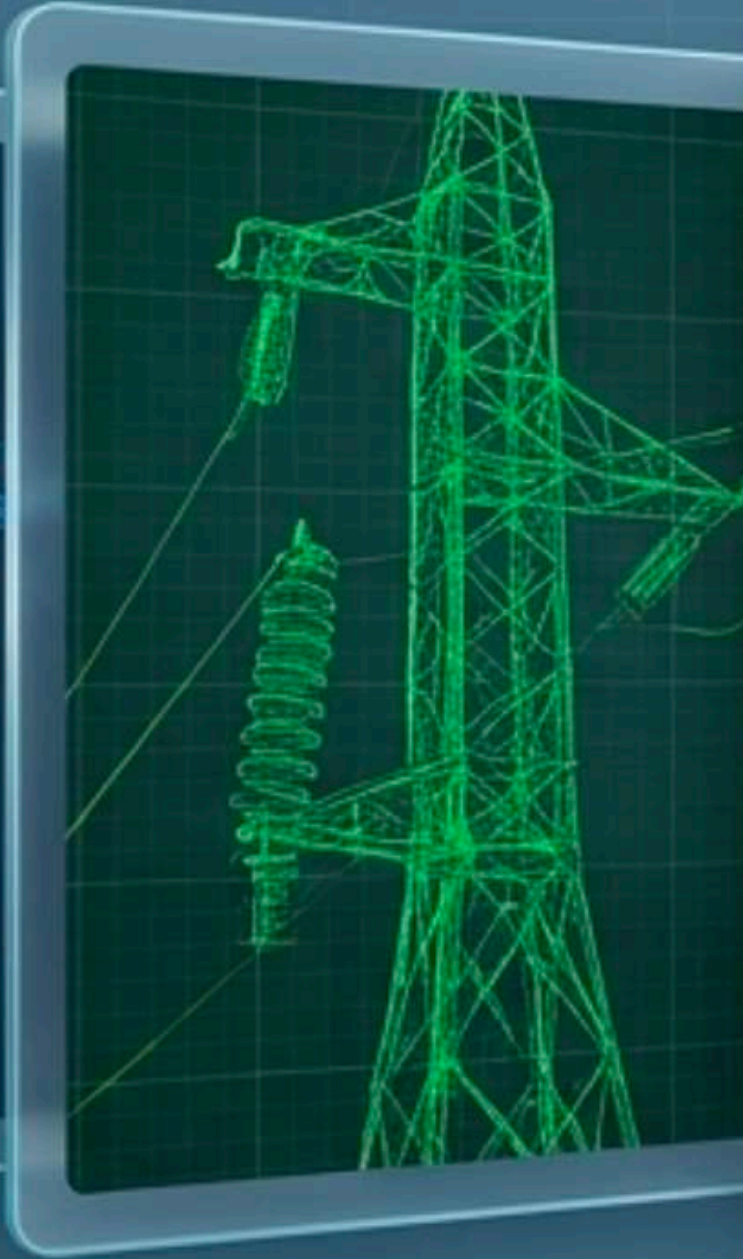
Automatic return

Obstacle avoidance
(trees, buildings, wires)

Without human intervention.
Stable data quality.



Smart power line inspection: AI + Computer Vision



Recognition of towers,
wires, insulators

Auto-zoom and
focus on objects

Detection of
defects and risks

Real-time analysis



Up to **80%** of network elements
are classified automatically

Intelligent Camera Lineup for Power Line Monitoring

ZONE 1: The Catalog

Standard and Advanced Solutions



Fixed Camera

Basic continuous monitoring of a designated network segment.



Multi-Lens Camera

360° panoramic coverage — zero blind spots.



PTZ Camera

Controllable pan/tilt and powerful optical zoom for detailed inspection of vast areas.



Thermal Imaging Camera

High-precision detection of overheating, temperature anomalies, and fault prevention.

ZONE 2: The Hero Product

Direct-to-Wire Camera

Our flagship autonomous solution for hard-to-reach segments.



Live-line installation:
setup without de-energizing the line.



Absolute control:
continuous real-time monitoring of power line conditions.



Complete autonomy:
inductive power supply directly from the phase conductor.



Smart analytics: built-in artificial intelligence and integrated sensors.



ZONE 3: The Ecosystem

Modular Integrations & Smart Sensors:



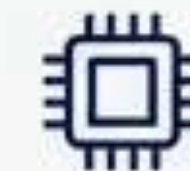
LiDAR — 3D clearance measurements and scanning of encroaching vegetation.



Micro weather station — localized monitoring of weather conditions.



Icing and vibration sensors — early warning for physical stress on the grid.



Edge AI — on-device peripheral data processing for instant alerts.

Platform architecture: From data collection to defect elimination

2. Data transmission

Online integration

Seamless transmission of telemetry and media via 4G/5G networks into a unified system.

3. AI Analysis

Computer Vision

Auto-recognition of defects (corrosion, breaks, overheating) with 90%+ model accuracy.

4. Visualization

Digital Twin

Building accurate 2D/3D models, creating a digital twin of the network, and analytics.

1. Data collection

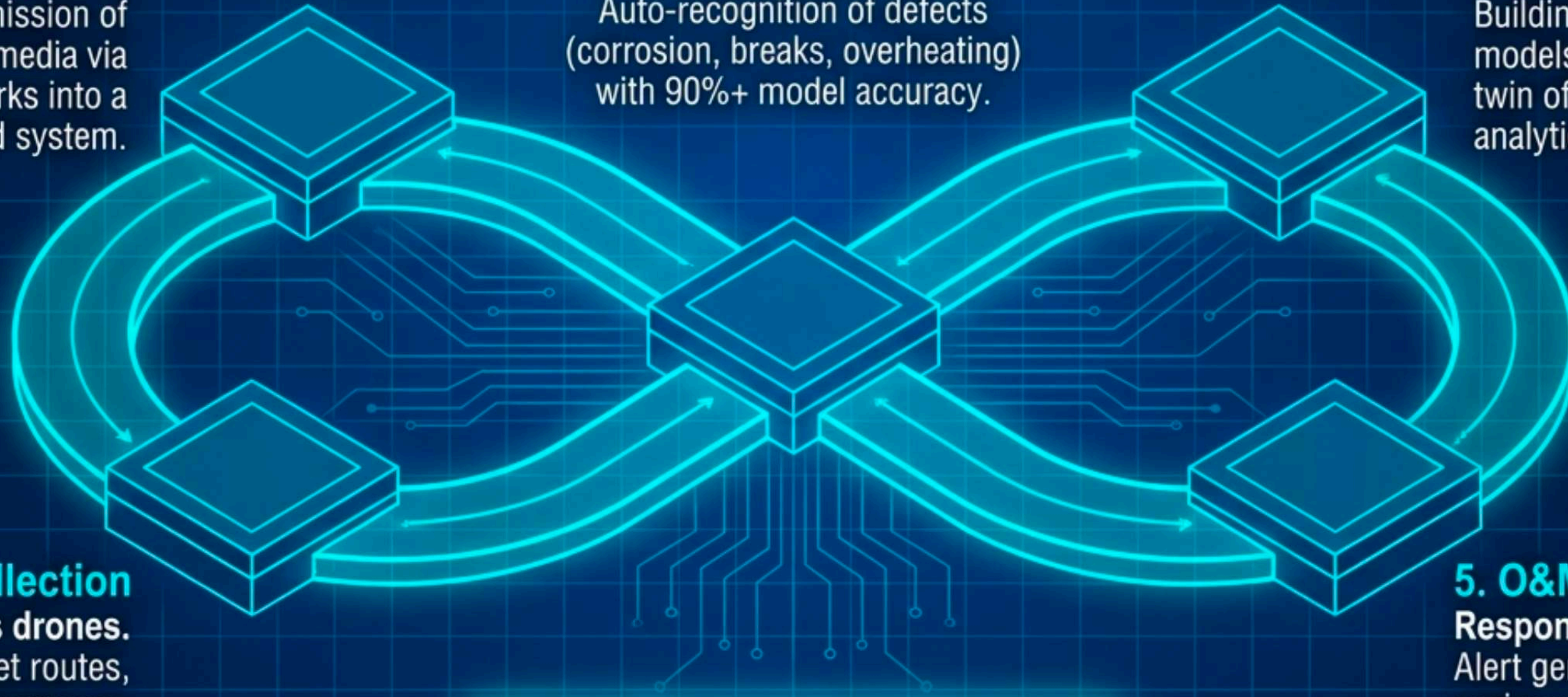
Autonomous drones.

Flights along preset routes, photo/video, IR cameras, LIDAR, and sensors.

5. O&M Management

Response.

Alert generation, automatic assignment of tasks to crews, and execution control.



Closed-loop management:
Everything in a single digital loop.

Power line flights: Intelligent navigation

01.

Flight along lines
(imitation flight)

02.

Automatic direction
detection

03.

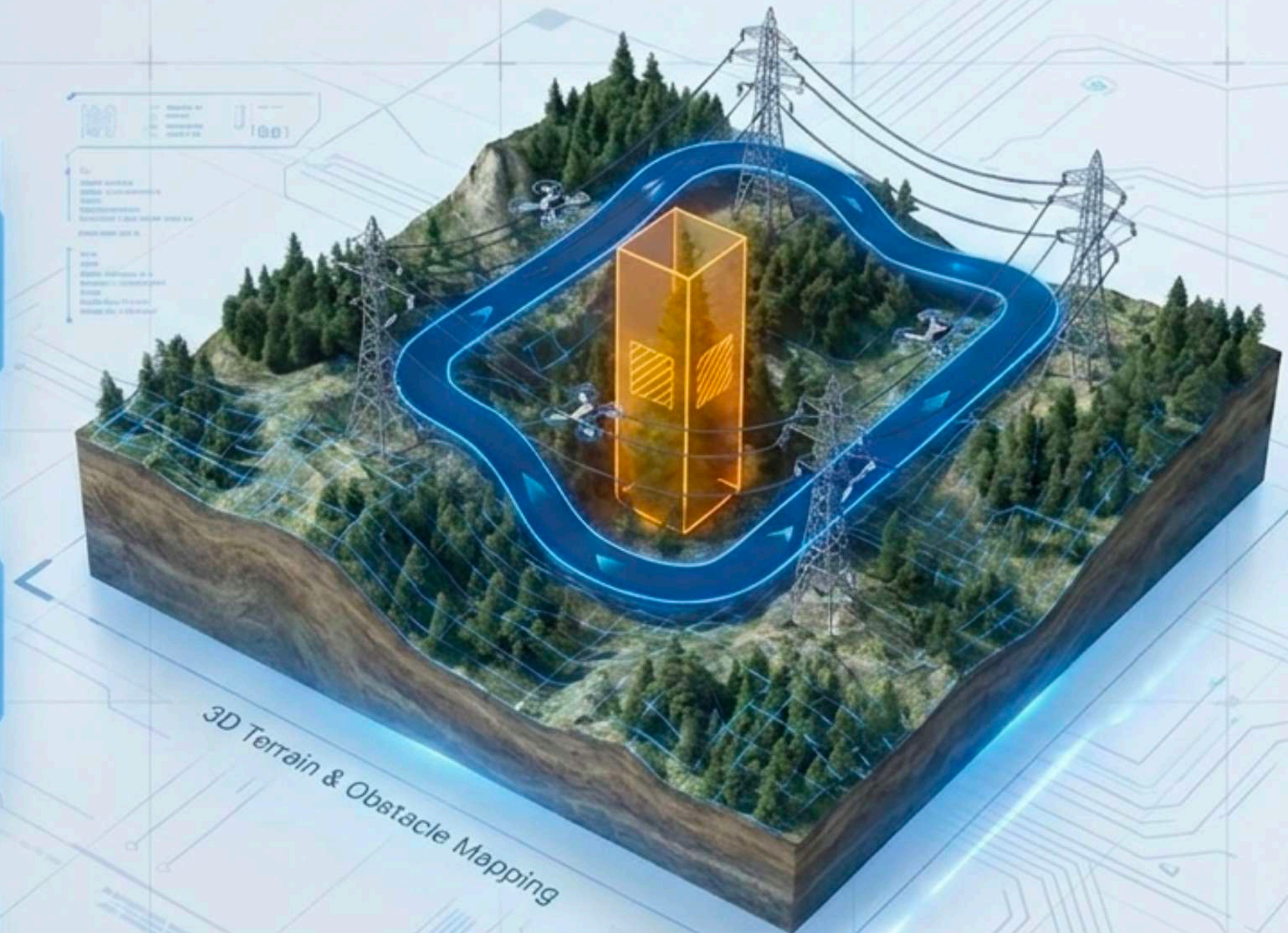
Route planning

04.

Bypassing
intersections and
obstacles



The drone itself “understands” the network structure



Data and analytics: Data-driven management



Geotagging photos



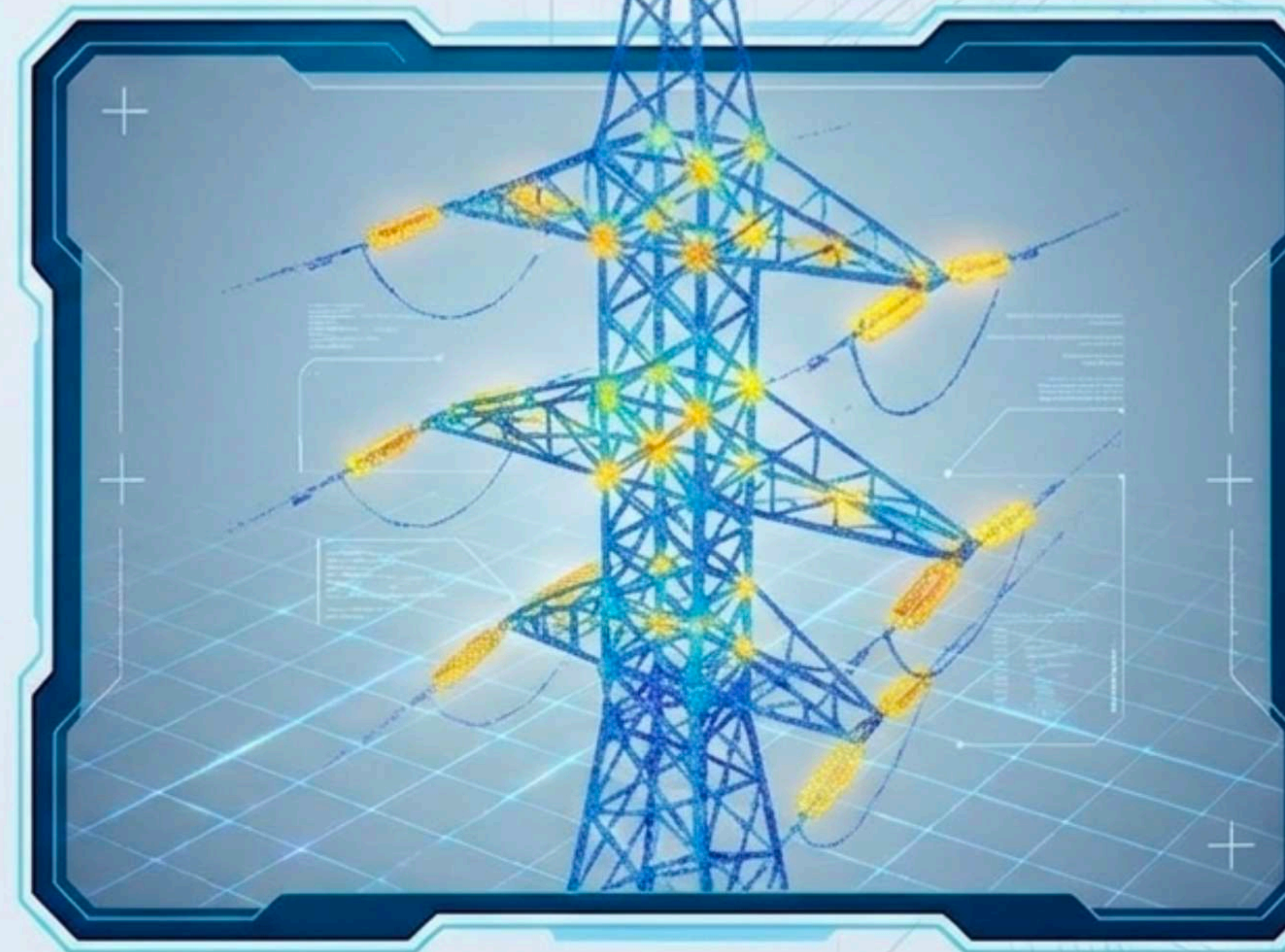
Online data transmission



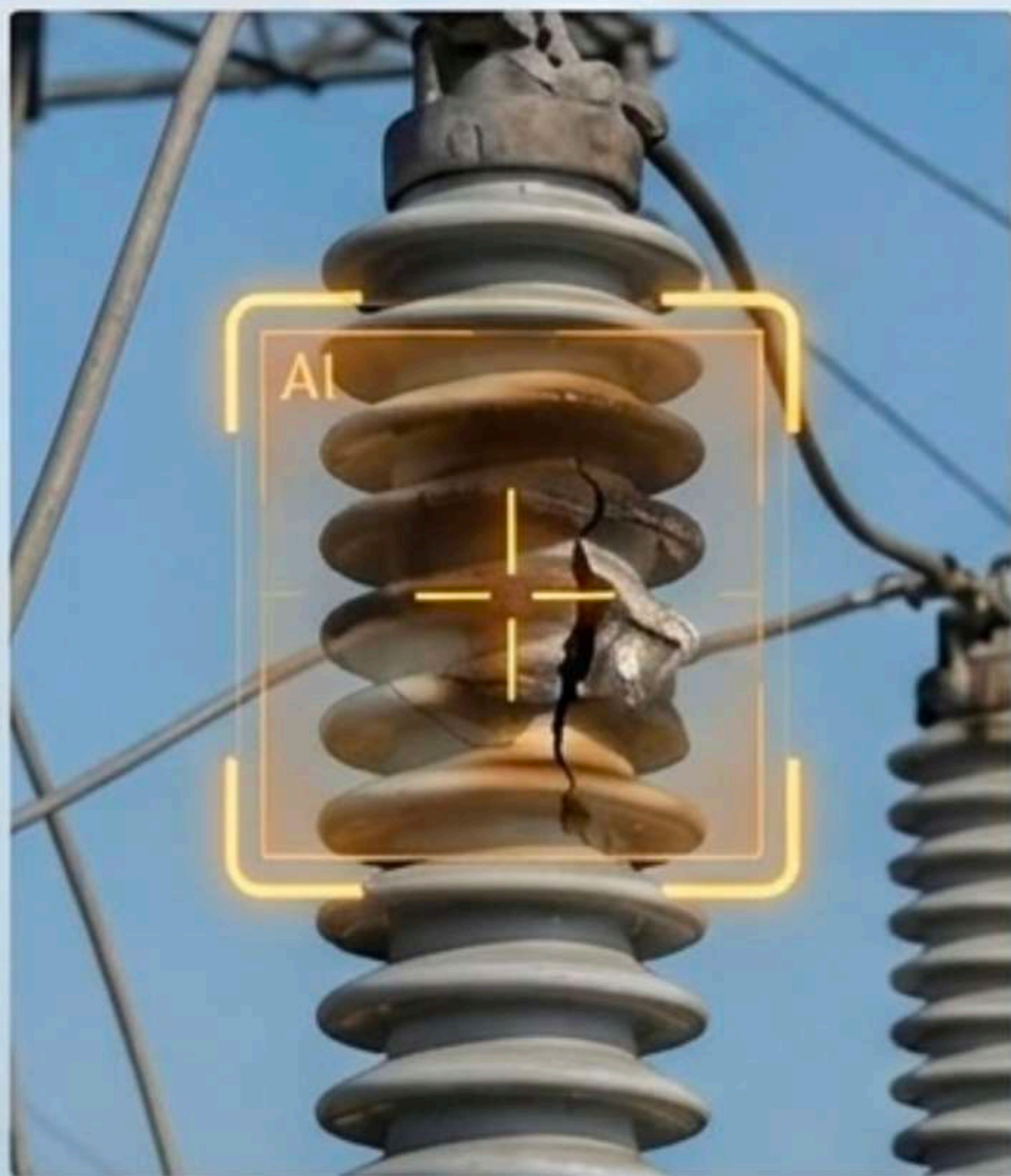
Inspection history



Condition monitoring
of each tower



Task management (O&M): From defect to task



Automatic task creation



Defect prioritization



Crew assignment and
execution control

Technologies: Key capabilities



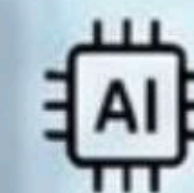
Obstacle
detection
(trees, buildings)



Real distance
to objects



Image quality
enhancement



AI lighting
correction



High accuracy even
in complex conditions

AI Diagnostics: Detected threats and problems solved



Solved problems

Personnel risks

Working at heights, under voltage, in hazardous areas.

Complex environment

Vast distances, mountains, snow, wind, heat.

Blind spots

Lack of real-time monitoring and a unified picture of the network state.

Detected defects

Hardware defects

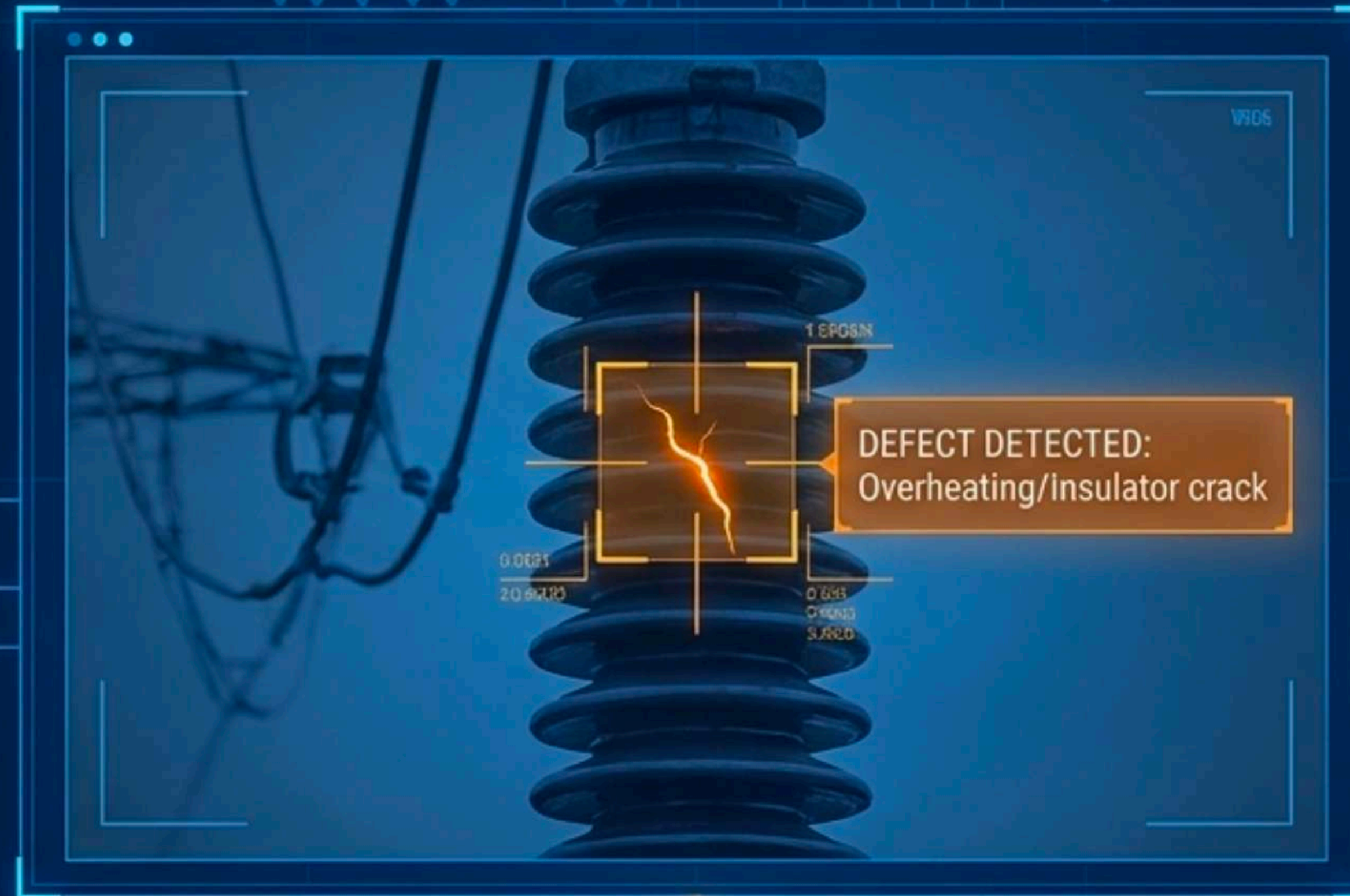
Corrosion, wire breaks, destruction/cracks in insulators, overheating of connections.

External threats

Foreign objects on lines, illegal construction in the power line protection zone.

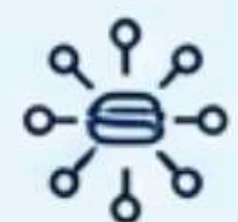
Environment

Encroaching vegetation, bird nests, consequences of forest fires.



Up to 80% of network elements are classified by the platform fully automatically.

Scalability: From pilot to the entire network



Fast connection of new lines



Centralized management



Support for thousands of kilometers of power lines



Business Impact: Transformation of the operating model



Traditional approach	Platform
High operating costs	↓ up to 60% reduction in O&M costs
Weeks of manual patrols	↑ Inspection speed increased by 10x
Relies on human factor	↑ AI accuracy up to 95%
High risks for personnel	✓ 0 risks to people



Reliability

Accident rate reduction up to -40%. Early defect detection and incident prevention.



Cost reduction

Elimination of expensive manual inspections and helicopters. Optimization of headcount.



Efficiency growth

Coverage of 100-500 km of lines per day. Full automation of routine checks.



Transparency and control

Unified monitoring platform. Online access to defect history and the digital twin.



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Power line digital twin + AI + drones = new management model

- Fewer people
- More data
- Faster decisions
- Higher safety



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